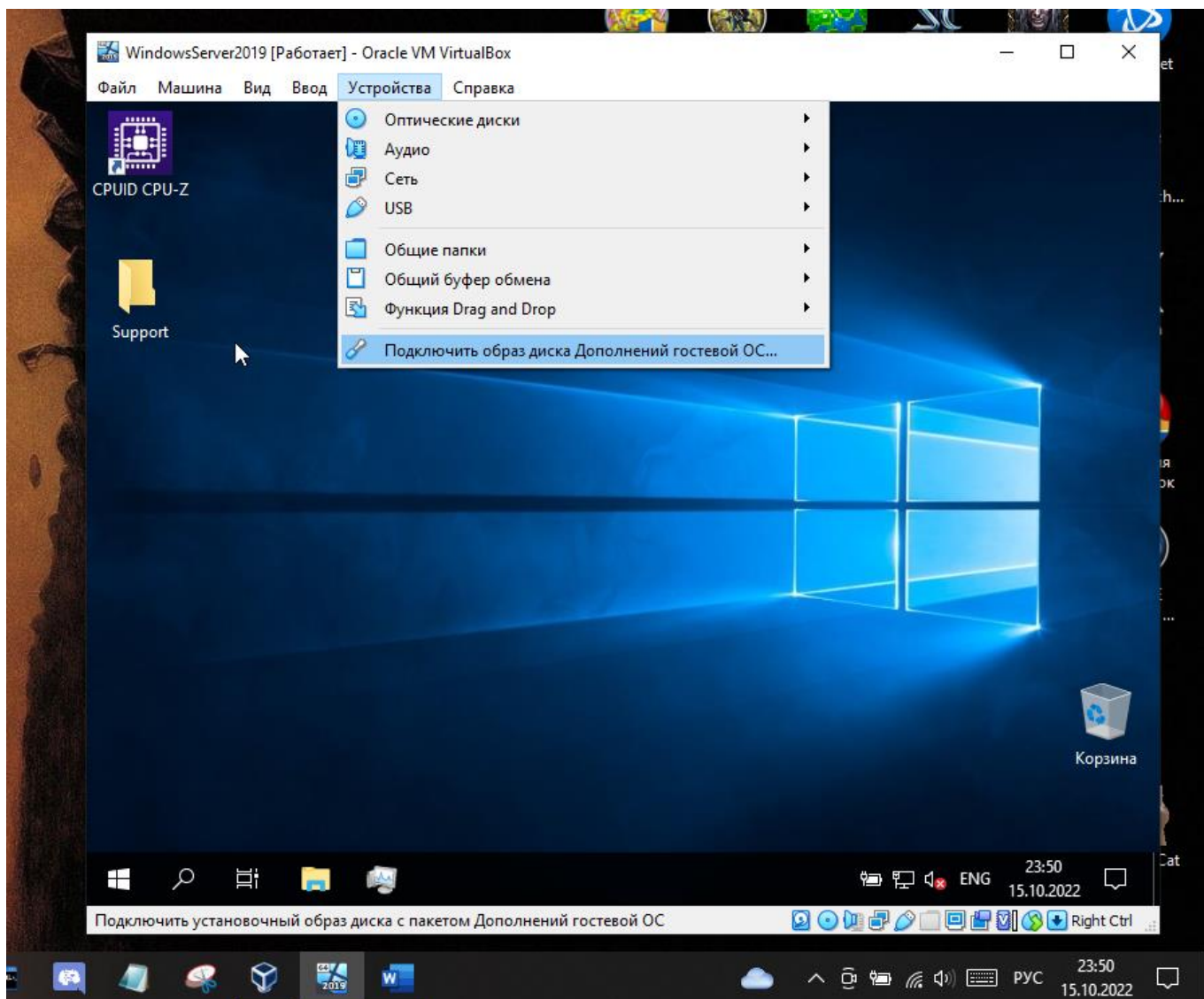


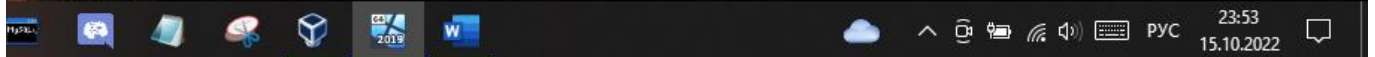
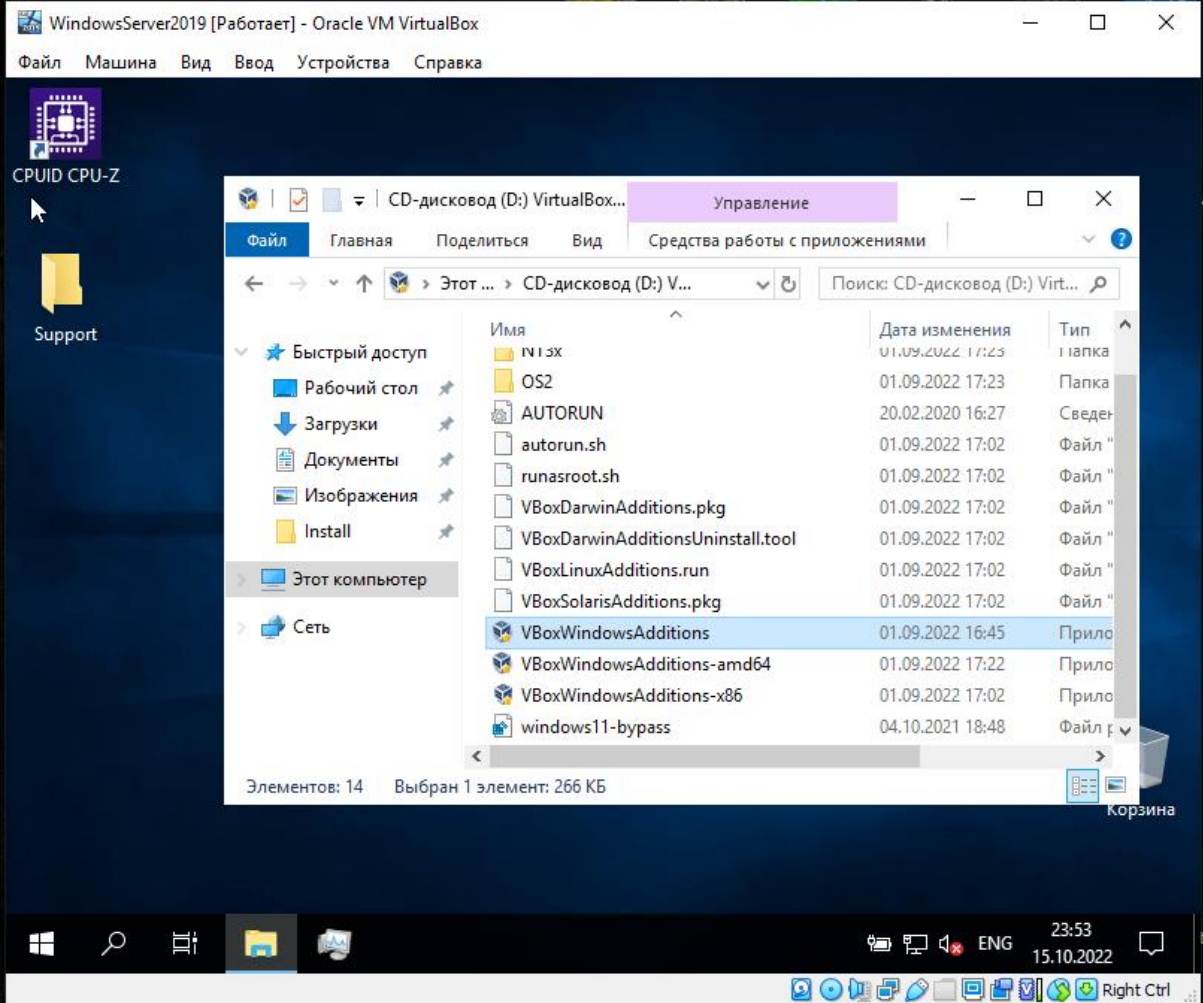
Цель работы: научиться настраивать сетевые адаптеры и подключать к ЛВС.

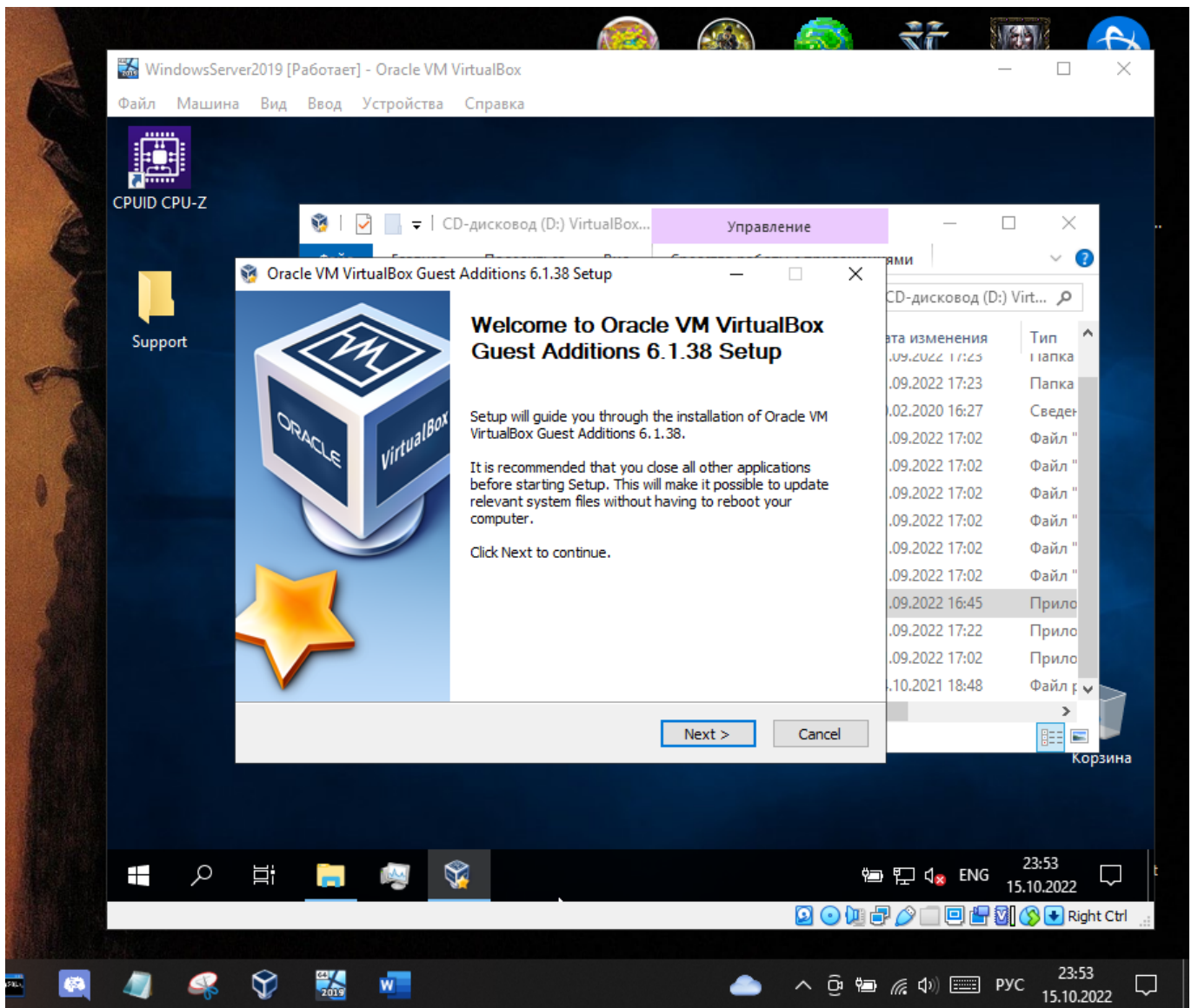
Выполнение заданий:

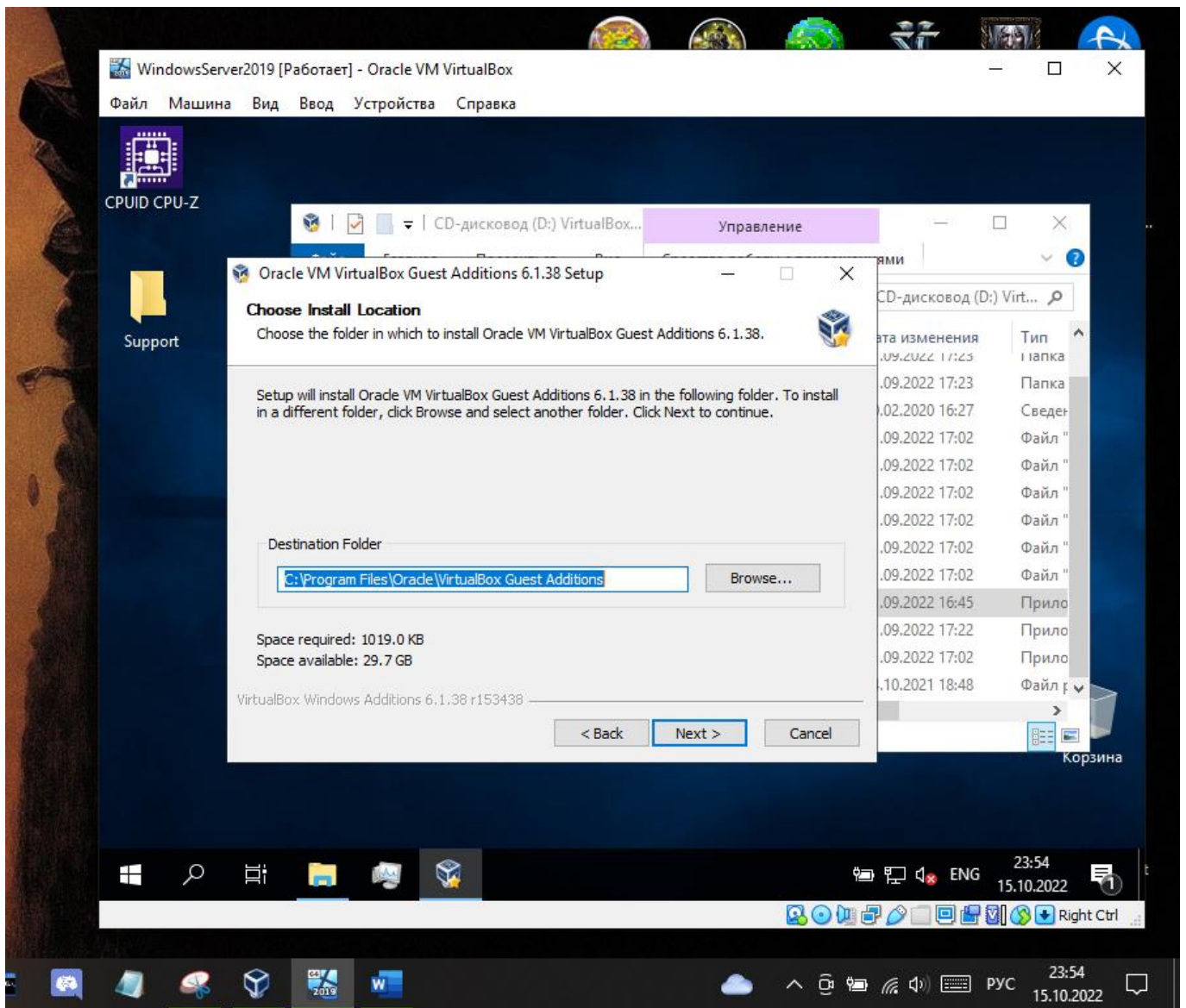
Вариант 11

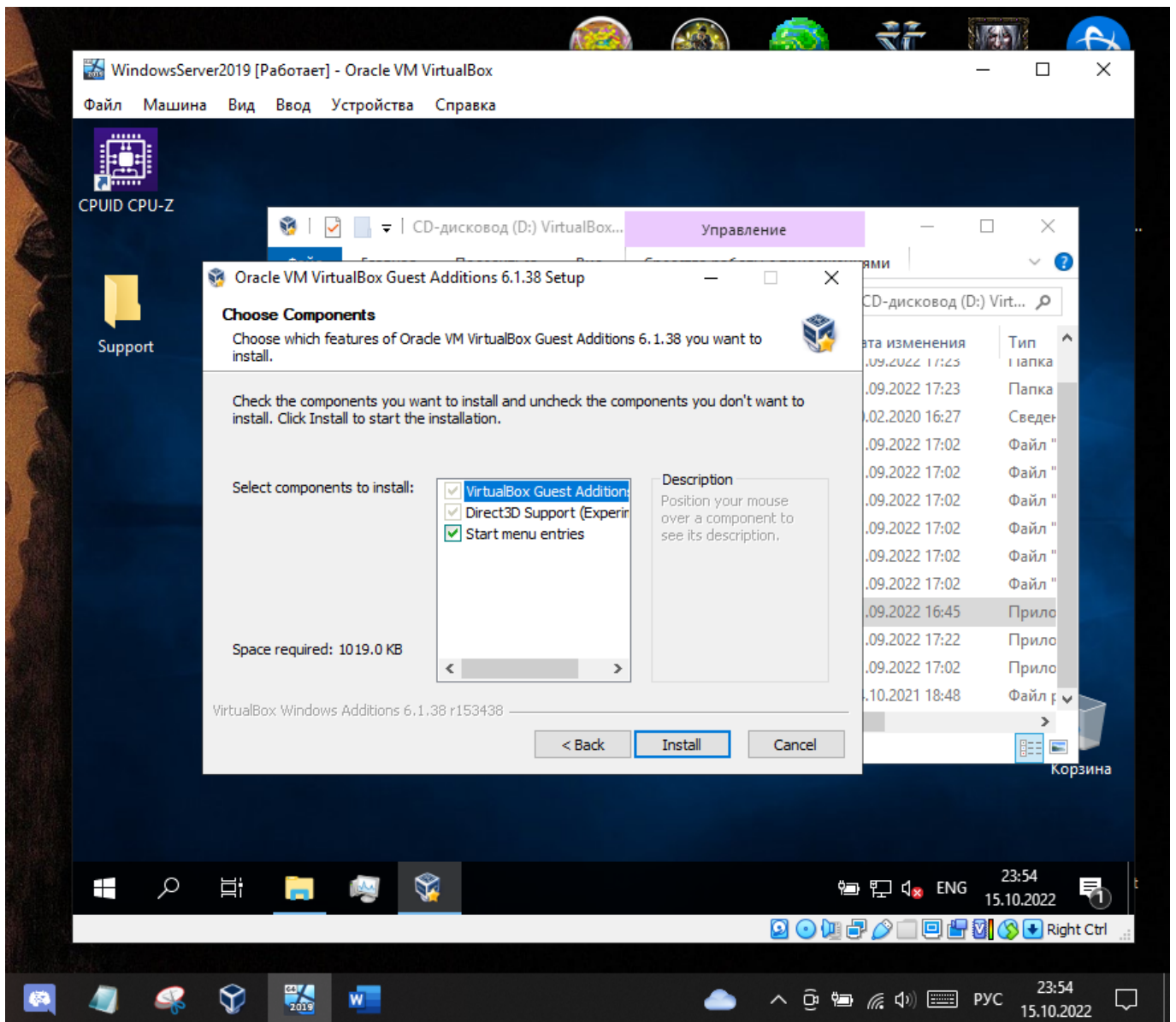
Общая папка

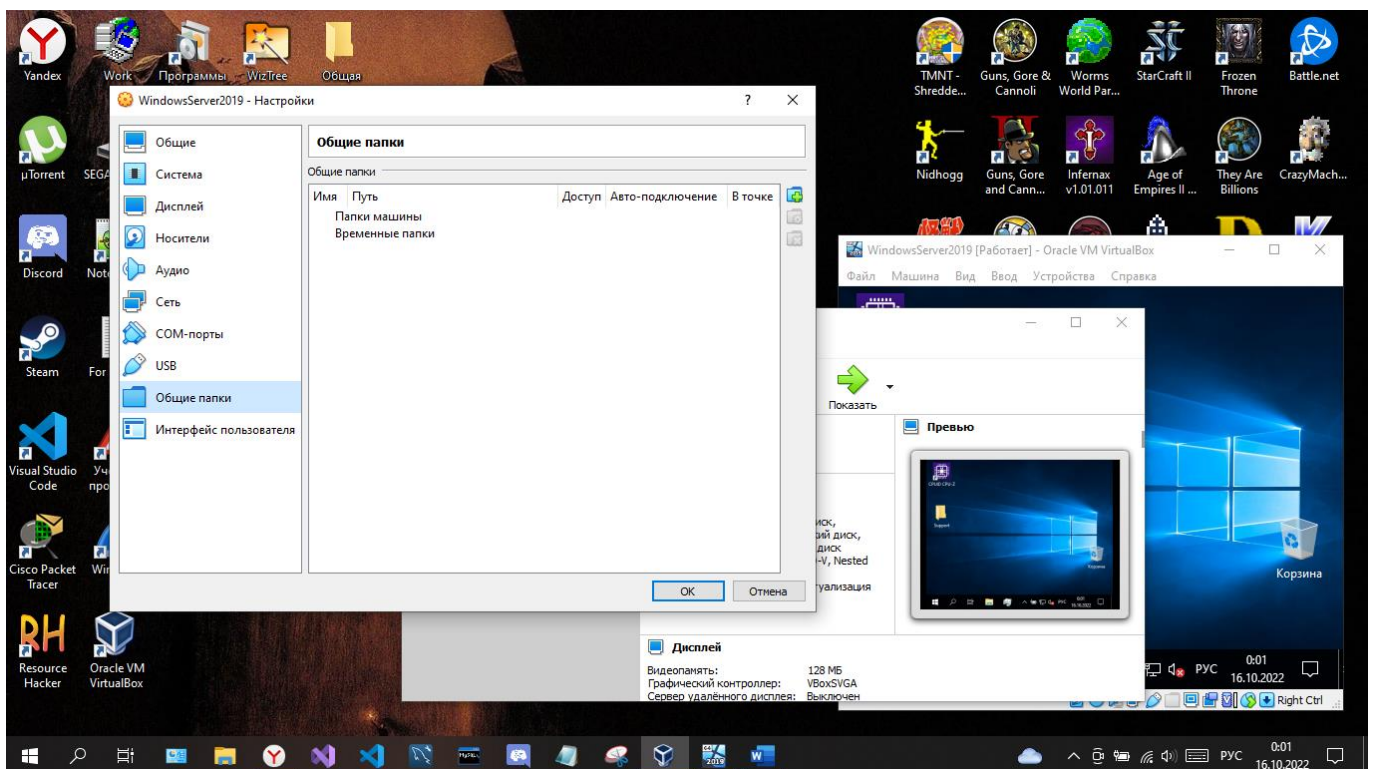
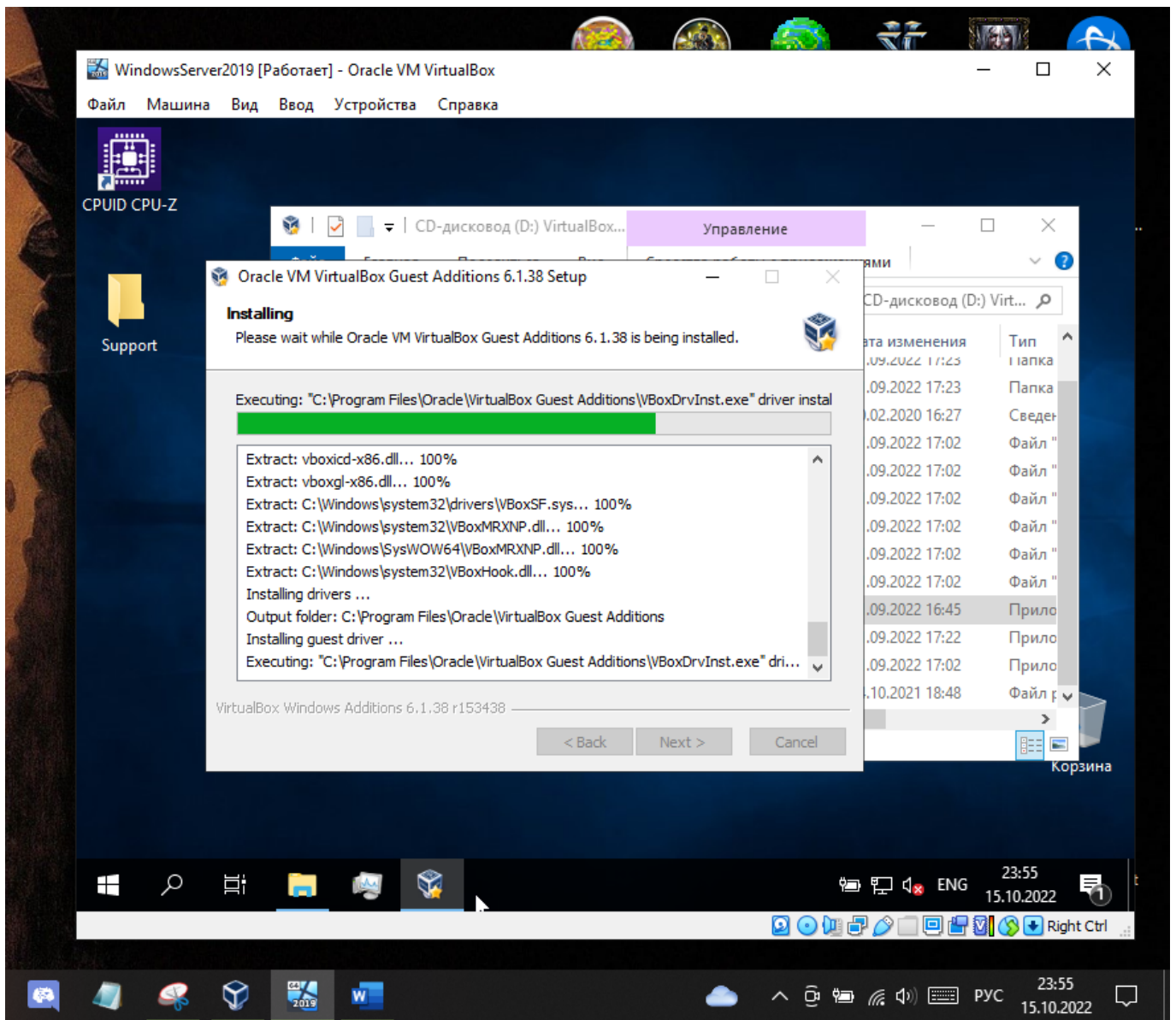


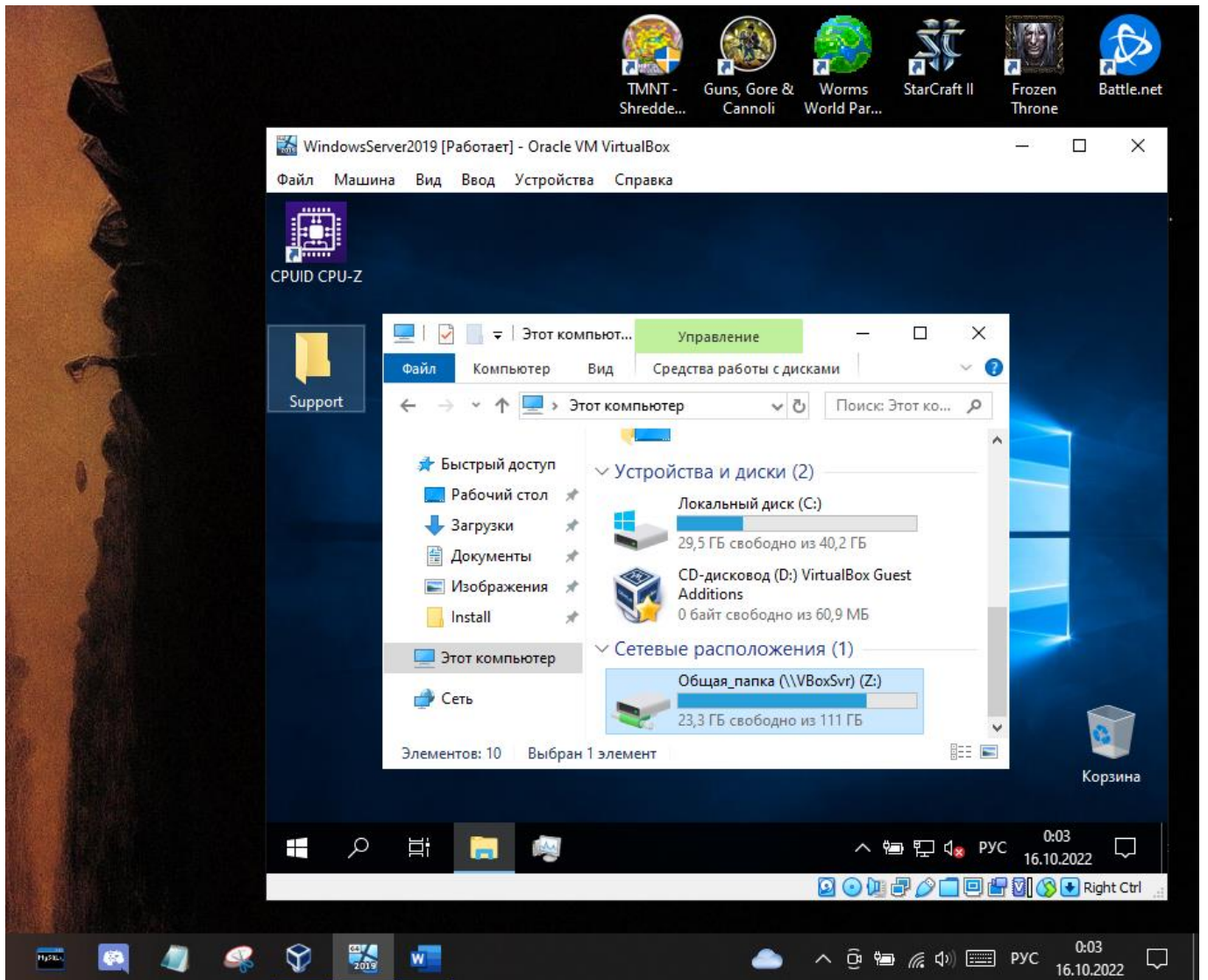
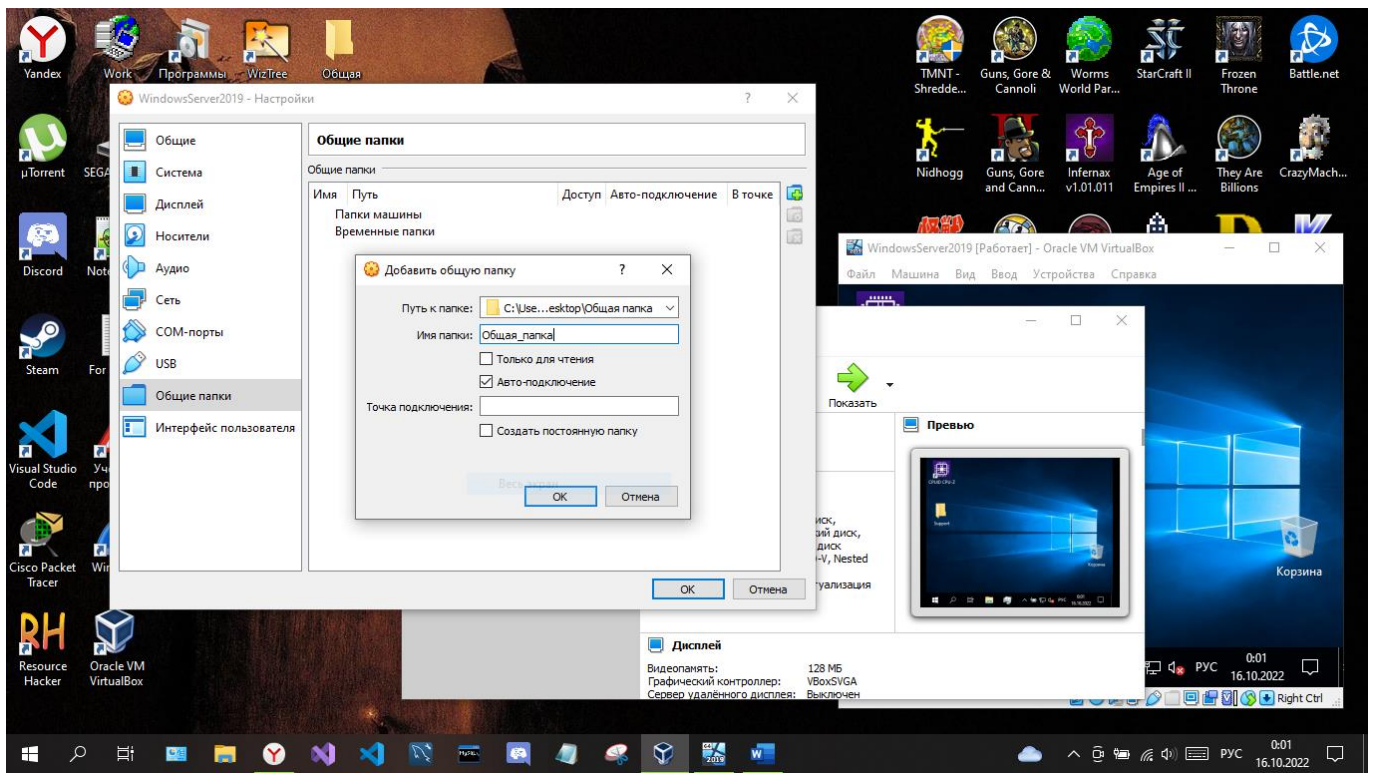


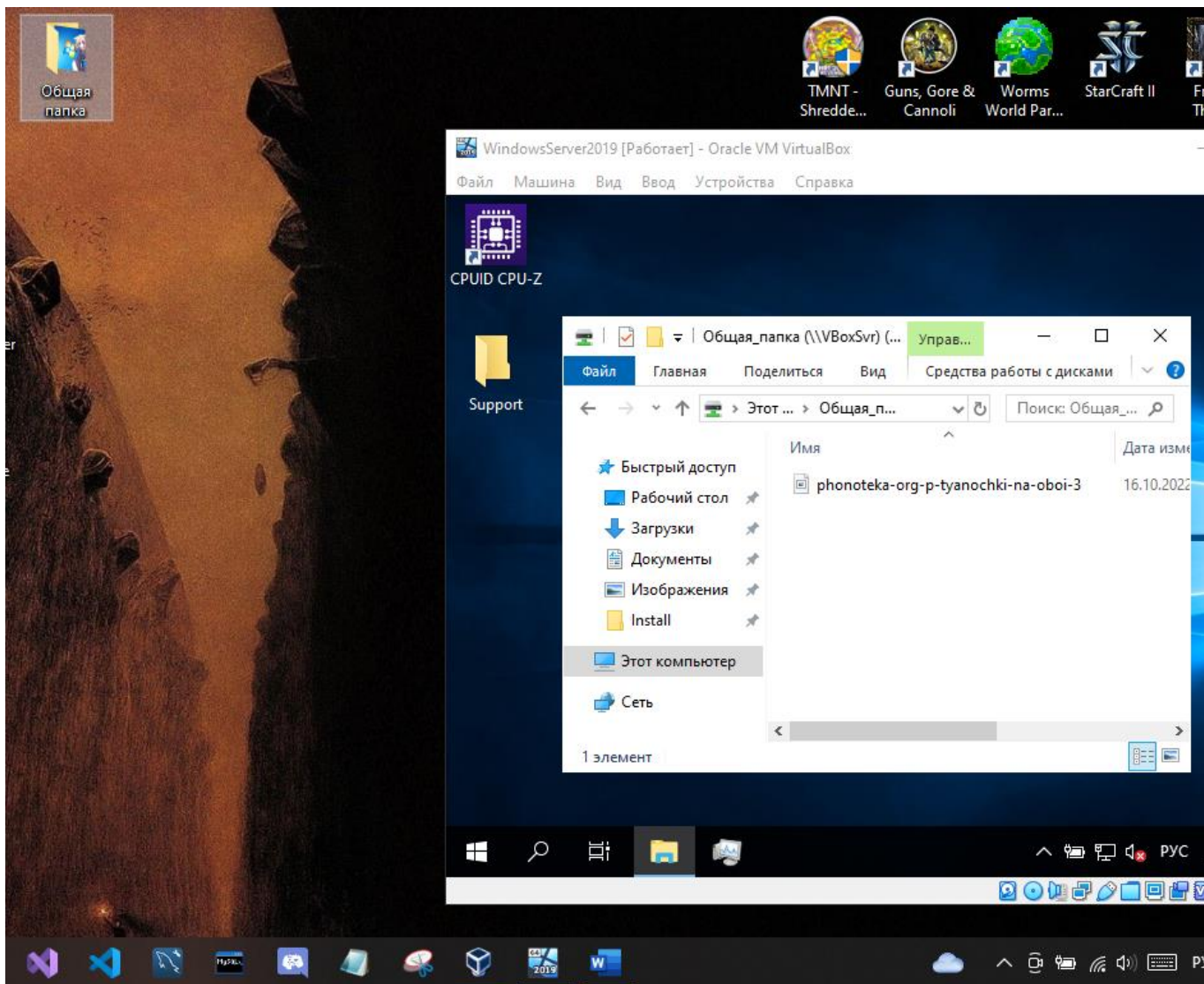


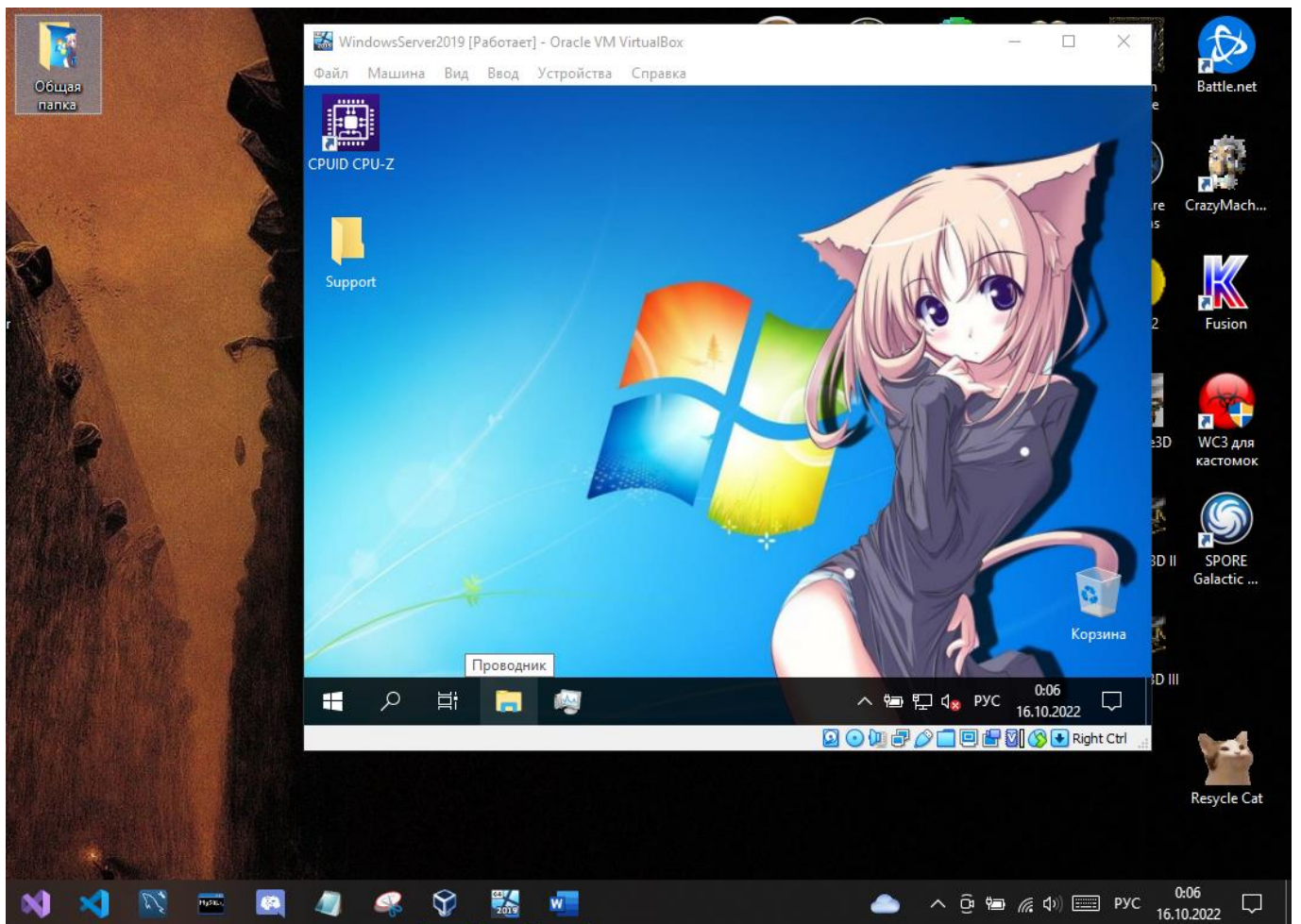












STP

Network Topology:

```

graph TD
    S0[Switch0] --- S1[Switch1]
    S1 --- S2[Switch2]
    S0 --- S2
  
```

PDU Information at Device: Switch0

OSI Model	Inbound PDU Details	Outbound PDU Details
At Device: Switch0 Source: Switch1 Destination: STP Multicast Address	In Layers Layer7 Layer6 Layer5 Layer4 Layer3 Layer2: IEEE 802.3 Header 00E0.F7E9.1B01 >> 0180.C200.0000 LLC STP BPDU Layer 1: Port FastEthernet0/1	Out Layers Layer7 Layer6 Layer5 Layer4 Layer3 Layer2: IEEE 802.3 Header 0060.3E98.6C03 >> 0180.C200.0000 LLC STP BPDU Layer 1: Port(s): FastEthernet0/3

1. FastEthernet0/1 receives the frame.

Time: 00:04:45.623

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 434, y: 287

Time: 00:03:05

Switch2

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, change
%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, change

Switch>en
Switch#show spa
Switch#show spanning-tree
VLAN0001

Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0002.1645.0A8D
This bridge is the root
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0002.1645.0A8D
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Fa0/2	Desg	FWD	19	128.2	P2p	
Fa0/1	Desg	FWD	19	128.1	P2p	

Switch#

Copper Cross-Over

Cisco Packet Tracer

File Edit Options View Tools Extensions Window Help

Logical Physical x: 434, y: 287

Time: 00:04:26

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, change
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, change

Switch>en
Switch#show spa
Switch#show spanning-tree
VLAN0001

Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0002.1645.0A8D
Cost 19
Port 2 (FastEthernet0/2)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 0010.1177.DB18
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Fa0/1	Desg	FWD	19	128.1	P2p	
Fa0/2	Root	FWD	19	128.2	P2p	

Switch#

Switch1

Physical Config CLI Attributes

IOS Command Line Interface

Press RETURN to get started!

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, change
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, change

Switch>en
Switch#show spa
Switch#show spanning-tree
VLAN0001

Spanning tree enabled protocol ieee
Root ID Priority 32769
Address 0002.1645.0A8D
Cost 19
Port 2 (FastEthernet0/2)
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)
Address 00D0.BAA9.11B8
Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec
Aging Time 20

Interface	Role	Sts	Cost	Prio.	Nbr	Type
Fa0/2	Root	FWD	19	128.2	P2p	
Fa0/1	Altn	BLK	19	128.1	P2p	

Switch#

Copper Cross-Over

Cisco Packet Tracer

File Edit View Help

Logical

Switch0

Physical Config CLI Attributes

IOS Command Line Interface

```
Switch#show spanning-tree
VLAN0001
  Spanning tree enabled protocol ieee
  Root ID    Priority    32769
            Address     0002.1645.0A8D
            Cost        19
            Port        2(FastEthernet0/2)
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec

  Bridge ID  Priority    32769  (priority 32768 sys-id-ext 1)
            Address     0010.1177.DB18
            Hello Time  2 sec  Max Age 20 sec  Forward Delay 15 sec
            Aging Time  20

Interface           Role Sts Cost      Prio.Nbr Type
-----
Fa0/1                Desg FWD 19        128.1    P2p
Fa0/2                Root FWD 19        128.2    P2p

Switch#conf t
Enter configuration commands, one per line.  End with CNTL/Z.
Switch(config)#int fa0/2
Switch(config-if)#sh

Switch(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/2, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/2, changed state to down

Switch(config-if)#exit
Switch(config)#exit
Switch#
%SYS-5-CONFIG_I: Configured from console by console

Switch#
```

Copy Pa

Top

Copper Cross-Over

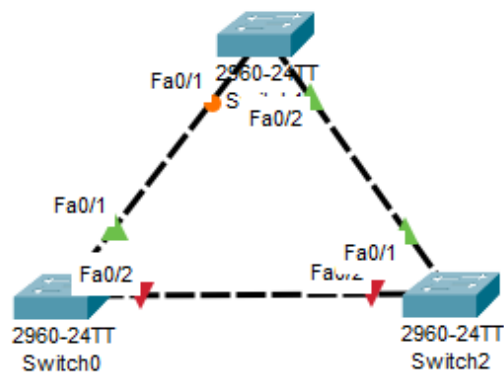
Time: 00:05:20

Windows taskbar icons: File Explorer, Edge, Word, PowerPoint, Teams, OneDrive, etc.



Logical Physical x: 783, y: 171

Root

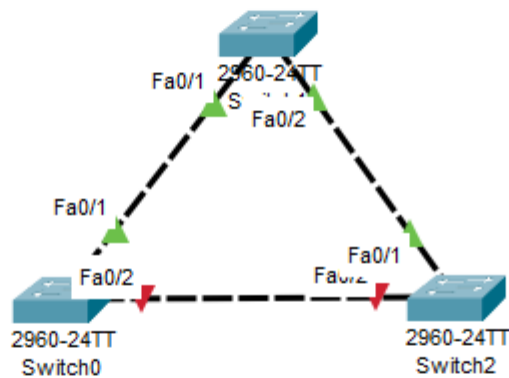


Time: 00:05:52



Copper Cross-Over



**Logical** **Physical** x: 574, y: 116

Time: 00:06:39



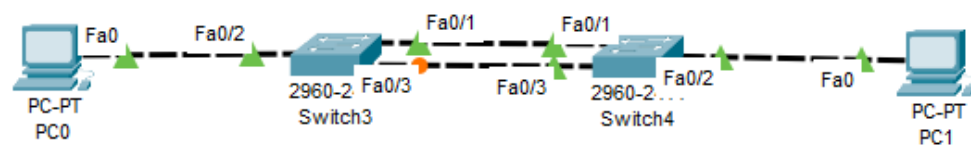
Copper Cross-Over





Logical Physical x: 670, y: 359

Root



Time: 00:04:08

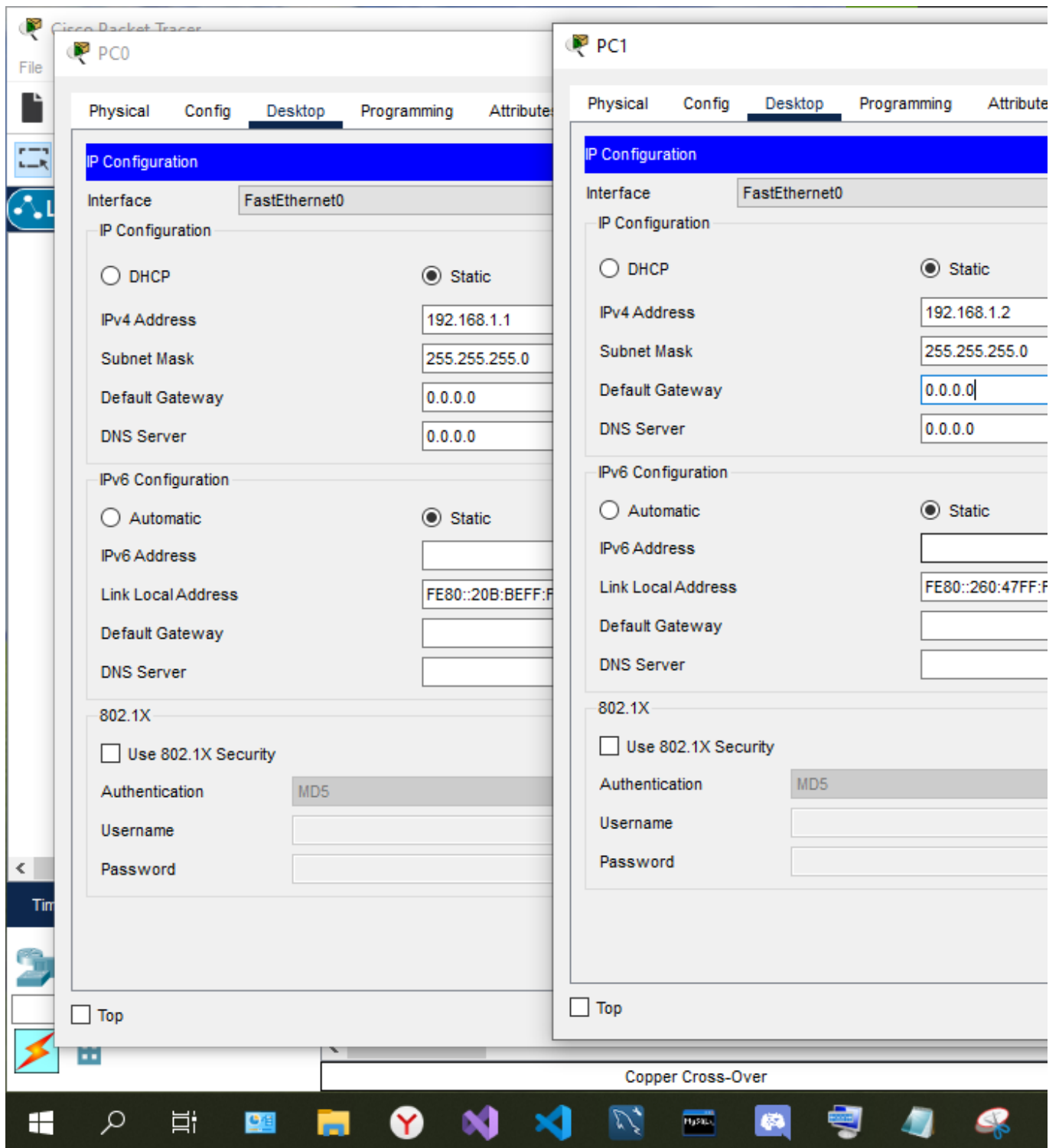


Realtime



Copper Cross-Over





PC1

Physical Config **Desktop** Programming Attributes

Command Prompt

```
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128
Reply from 192.168.1.1: bytes=32 time=1ms TTL=128
Reply from 192.168.1.1: bytes=32 time<1ms TTL=128

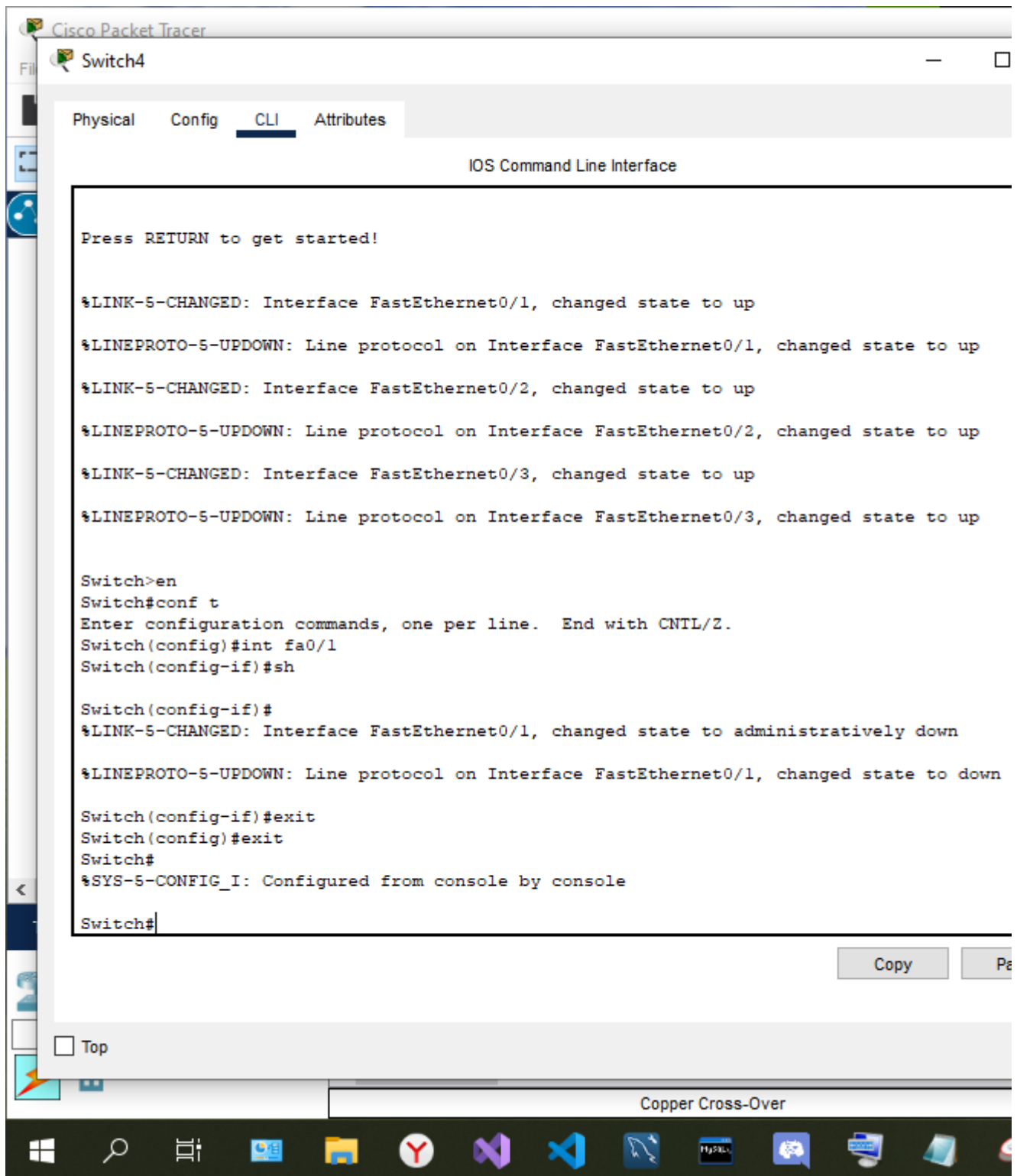
Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 1ms, Average = 0ms

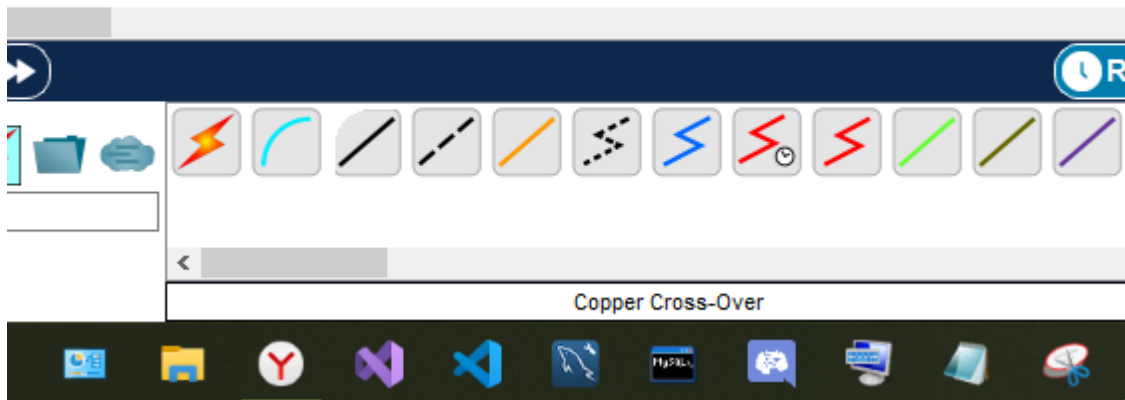
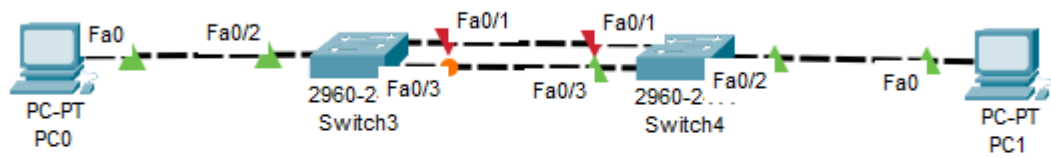
C:\>
```

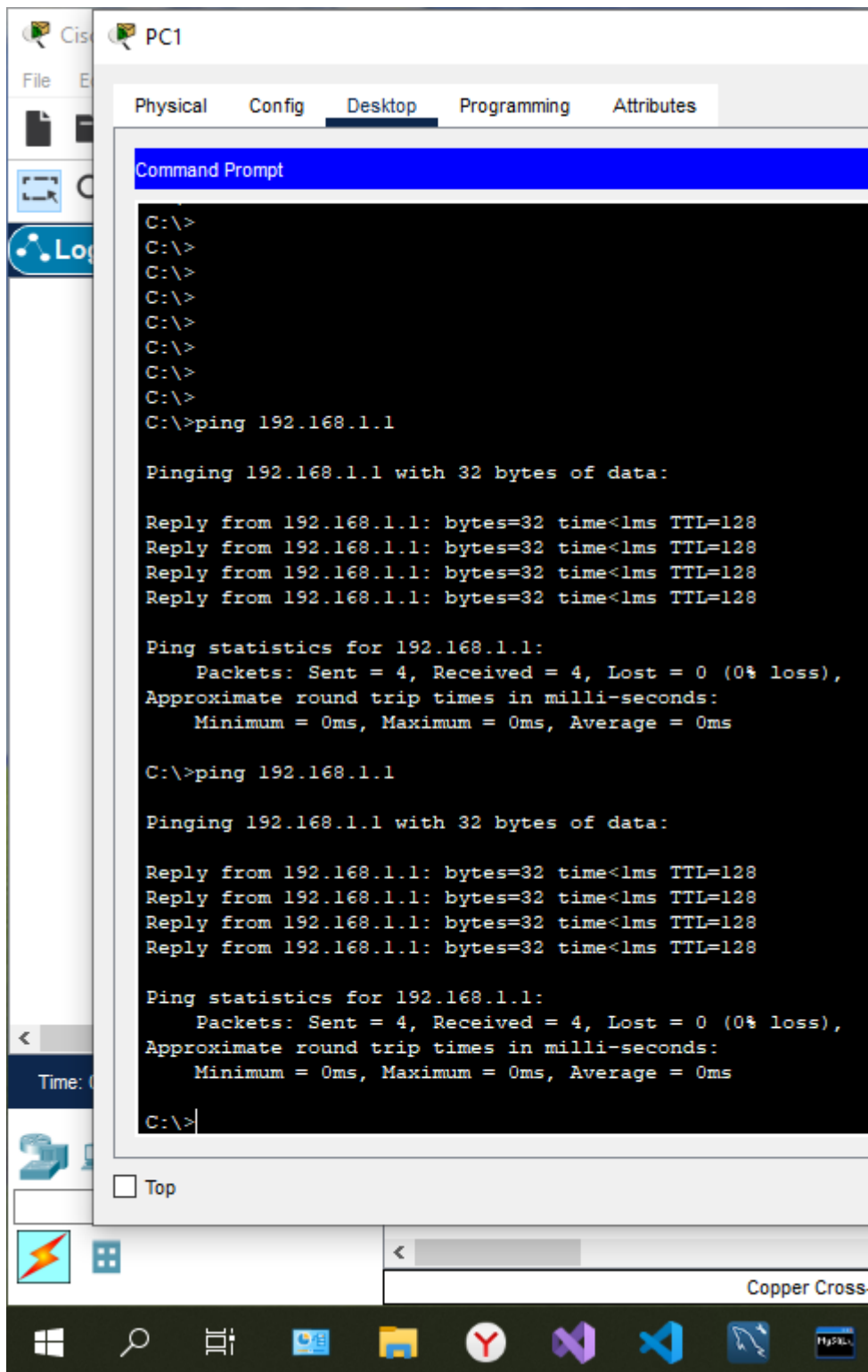
☐ Top

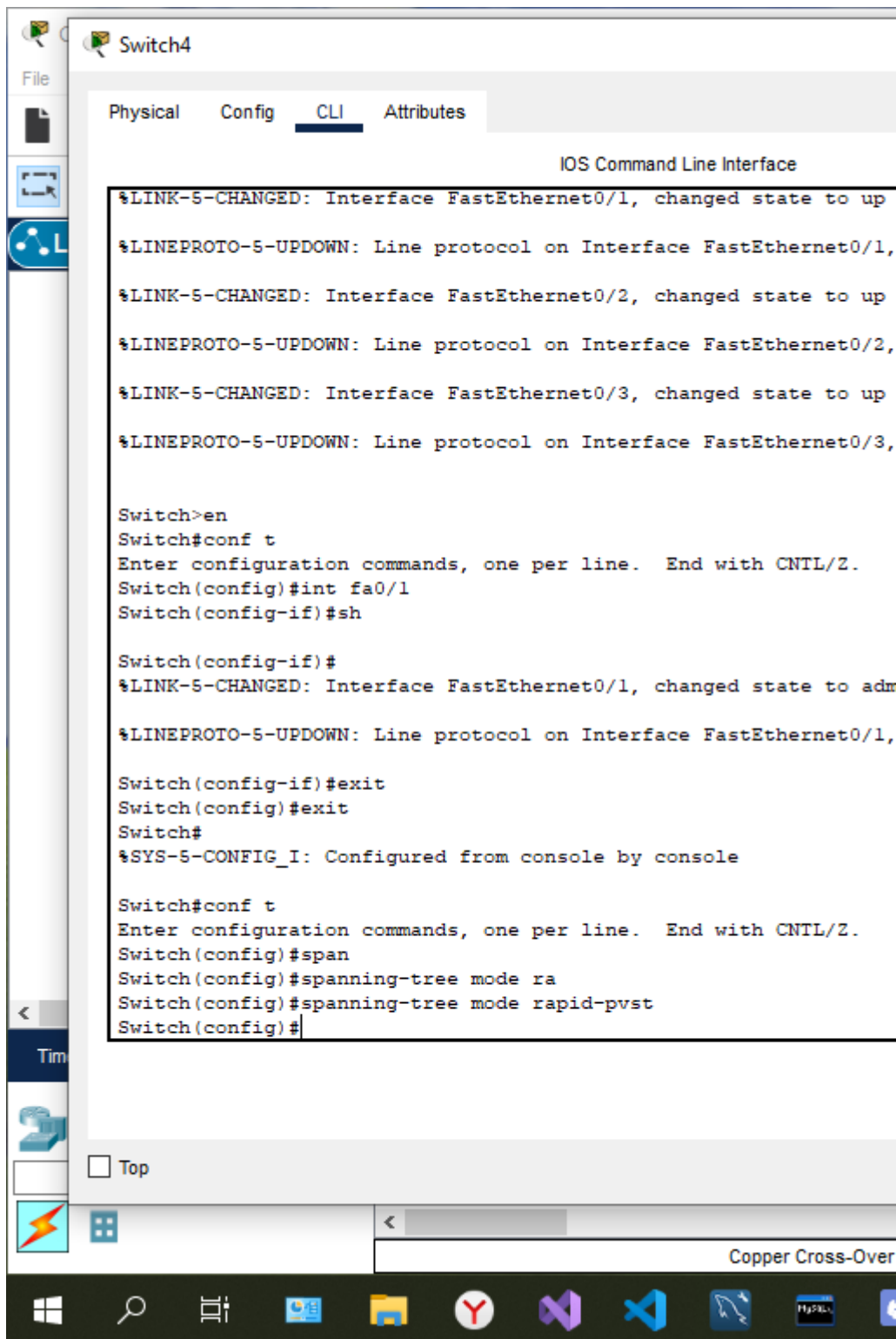
Copper Cros











Switch3

Physical

Config

CLI

Attributes

IOS Command Line Interface

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to down

Switch>en

Switch#conf t

Enter configuration commands, one per line. End with CNTL/Z.

Switch(config)#spa

Switch(config)#spanning-tree mode rap

Switch(config)#spanning-tree mode rapid-pvst

Switch(config)#end

Switch#

%SYS-5-CONFIG_I: Configured from console by console

Switch#show spann

Switch#show spanning-tree

VLAN0001

Spanning tree enabled protocol rstp

Root ID Priority 32769

Address 00D0.589E.EC9B

Cost 19

Port 3(FastEthernet0/3)

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Bridge ID Priority 32769 (priority 32768 sys-id-ext 1)

Address 00E0.B025.CC77

Hello Time 2 sec Max Age 20 sec Forward Delay 15 sec

Aging Time 20

Interface	Role	Sts	Cost	Prio.Nbr	Type
Fa0/3	Root	LRN	19	128.3	P2p
Fa0/2	Desg	FWD	19	128.2	P2p

Switch#

Copy

Paste

☐ Top

Copper Cross-Over

Switch(config-if)#

%LINK-5-CHANGED: Interface FastEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/1, changed state to up

Switch(config-if)#exit

Switch(config)#exit

Switch#

%SYS-5-CONFIG_I: Configured from console by console

Switch#

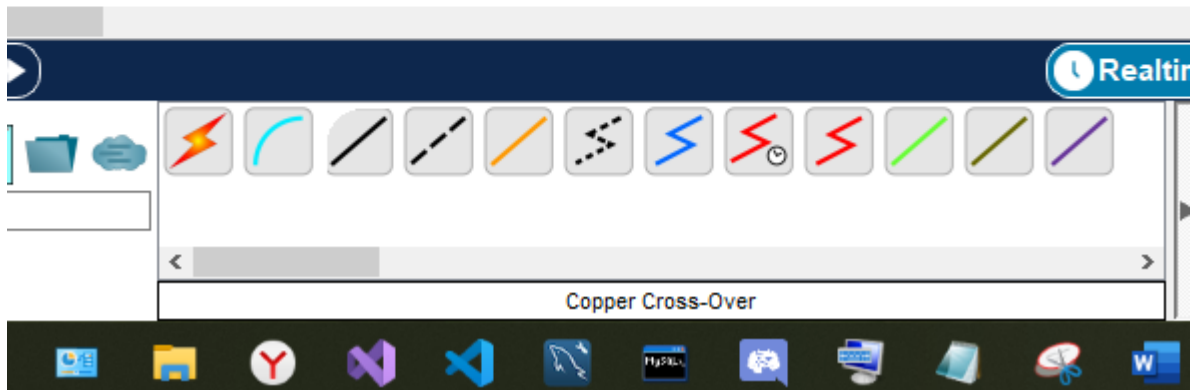
Copy

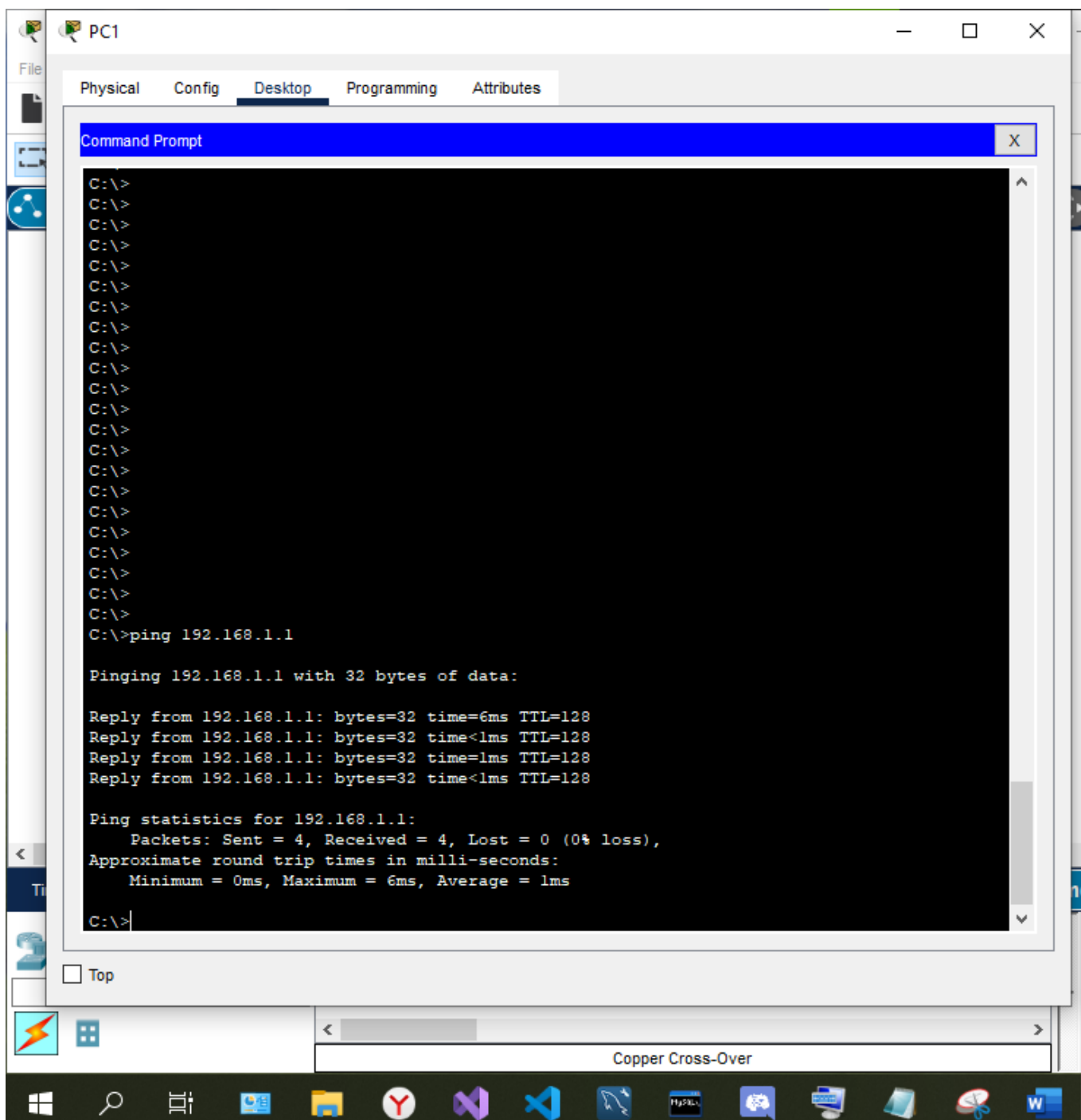
☐ Top

Copper Cross-Over

Windows Taskbar

Taskbar Icons





Контрольные вопросы

1. Какие этапы включает в себя монтаж ЛВС?
 - Осмотр объекта специалистом по установке ЛВС;
 - Подбор необходимого оборудования;
 - Прокладывание кабельных маршрутов;
 - Настройка и запуск оборудования.
2. Выделите основные функции локально-вычислительных путей
Основные функции локально-вычислительных путей:

- быстрая отправка электронной документации в различные филиалы компании;
- непрерывная коммуникация между сотрудниками компании, находящимися в центральном офисе и на удаленных объектах;
- предоставление доступа из разных точек сети к принтерам и факсам;
- возможность использовать архивы и иные хранилища информации одновременно несколькими сотрудниками компании;
- возможность совместного доступа в Интернет;
- обеспечение возможности удаленной коммуникации, в число которой входят различные конференции;
- функция объединения компьютеров, которые используют для работы разные операционные системы (Apple, Linux, Windows и т.д.).
- удаленное администрирование сети.

3. Перечислите наиболее распространенные технологии передачи информационных данных

Token Ring, Ethernet, FDDI, ArcNet, Ethetnet.

4. Как происходит установка кабельных маршрутов?

Прокладка кабельных маршрутов строго в соответствии с заранее разработанным проектом, при этом необходимо учитывать все особенности помещения, куда проводится кабельная трасса.

Вывод: я научился настраивать сетевые адаптеры и подключать к ЛВС.